SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	VAIBHAVA LAKSHMI K	**Reg. No.:**	192511310
Section / Batch:	2025-2029	**Date:**	

Code of Conduct

I VAIBHAVA LAKSHMI K (192511310) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: VAIBHAVA LAKSHMI K

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"The Turing Test … judges a machine to be intelligent if its internal reasoning always follows the laws of logic" and "thinking rationally." The Turing Test evaluates human-like behaviour, not whether internal reasoning follows formal logic.
S2	Incorrect	"AI winters … occurred in the 1950s, before the 1956 Dartmouth workshop" and "That workshop then revived the field." AI winters occurred later, during periods of reduced funding and interest.
S3	Incorrect	"A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance" and "the taxi … was not rational." Rationality does not require knowing future outcomes.
S4	Incorrect	The Performance measure and Actuators are interchanged. Steering, accelerator, brake and horn are actuators, whereas a safe, fast, legal and comfortable trip describes performance measures.
S5	Incorrect	"Partially observable, stochastic, episodic." Chess without a clock is fully observable, deterministic and sequential. Static and discrete are correct.
S6	Correct	No erroneous claim. A model-based reflex agent maintains an internal state and updates it using a model of how the environment changes and how its actions affect it.

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

S1 – Turing Test	The Turing Test belongs to the **acting humanly** approach. The student confused it with the **thinking rationally** approach, which focuses on reasoning according to logical rules.
S2 – AI winters	The student confused the beginning of AI as a field with later periods of reduced funding and interest. The term "artificial intelligence" was introduced at the 1956 Dartmouth workshop, but AI winters occurred in later decades.
S3 – Rationality and advance knowledge	Rationality was confused with omniscience. A rational agent selects an action based on the available percept sequence, prior knowledge, available actions and performance measure. It cannot always predict unexpected events.
S4 – PEAS	The four PEAS components were confused. Performance measure specifies success criteria, Environment identifies the surroundings, Actuators perform actions and Sensors collect information.
S5 – Chess environment	The student confused observability, determinism and task structure. The chessboard and pieces are visible, legal moves determine the resulting state, and each move affects later decisions. Without a clock, the environment does not change merely because time passes.
S6 – Model-based reflex agent	No error. The statement correctly explains how an internal state and a model help an agent act when its current percept does not reveal the complete environment state.

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1	The Turing Test, proposed by Alan Turing in 1950, evaluates whether a machine can exhibit behaviour indistinguishable from that of a human in a suitable conversation. It is an example of the **acting humanly** approach to AI.	The test evaluates observable behaviour rather than requiring the machine to follow formal logical reasoning internally.
S2	AI winters were periods of reduced funding, expectations and interest in AI, occurring in later decades, notably during the 1970s and again in the late 1980s. The 1956 Dartmouth workshop is associated with the introduction of the term "artificial intelligence" and the establishment of AI as a field.	The Dartmouth workshop preceded the later AI winters; it did not occur after them or revive the field following them.
S3	A rational agent chooses the action expected to maximise its performance measure, given its percept sequence, prior knowledge and available actions. It does not need to know the outcome of every action in advance. Therefore, choosing a route shown as clear by available traffic reports can be rational even if an unforeseeable landslide later blocks it.	Rationality is judged using the information available when the decision is made, not solely by the eventual outcome.
S4	**Performance measure:** safe, fast, legal and comfortable trip. **Environment:** roads, other traffic, pedestrians and weather. **Actuators:** steering, accelerator, brake and horn. **Sensors:** cameras, GPS and speedometer.	The corrected PEAS description assigns success criteria to Performance measure and physical controls to Actuators.
S5	Chess played without a clock is a **fully observable, deterministic, sequential, static and discrete** task environment. It is also competitive and multi-agent because two players make opposing decisions.	The board state is visible, moves have predictable rules, each move affects later moves, no clock-driven change occurs, and the game uses discrete states and actions.
S6	**No change.** A model-based reflex agent maintains an internal state and updates it using a model of how the world evolves and how its own actions affect it. This helps it act sensibly when the current percept does not reveal the complete state.	

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i) Was the taxi's choice rational? Explain using the four factors.

The taxi's choice was rational if it selected the highway route using the information available at that time. Rationality is determined by four factors:

1. **Performance measure:** The taxi aims to provide a safe, legal, comfortable and efficient journey.
2. **Percept sequence:** All available traffic reports indicated that the highway was clear.
3. **Prior knowledge:** The taxi used its available knowledge of roads, traffic and route conditions. The landslide was unforeseeable.
4. **Available actions:** The taxi selected the highway route from the routes available to it at the time.

Therefore, choosing the highway was rational based on the available evidence. The unexpected landslide does not automatically make the original decision irrational. However, once the obstruction is detected, the taxi should update its information and choose a safe alternative if one is available.

(ii) Name the task-environment properties that make perfect foresight impossible.

- **Partially observable:** Sensors and reports may not reveal every road condition or hidden hazard.
- **Stochastic:** Unexpected events, such as a landslide, can occur and change the outcome.
- **Dynamic:** Road and traffic conditions can change while the taxi is operating.
- **Continuous:** Speed, distance, time and steering are continuous variables.
- **Multi-agent:** Other drivers and road users also make decisions that affect the taxi.

These properties prevent the taxi from knowing every future event with certainty.

(iii) Give one situation in which a simple reflex taxi behaves poorly. What does a model-based reflex taxi maintain?

A simple reflex taxi may behave poorly when it sees a red traffic signal but does not consider that it has already stopped behind a vehicle. If it reacts only to the current percept without retaining relevant information, it may repeatedly apply the same action or fail to account for the surrounding traffic situation.

A **model-based reflex taxi maintains an internal state** containing relevant information about the environment that may not be available in the current percept alone. It updates this state using a model of how the environment changes and how its actions affect it. This enables more appropriate decisions in partially observable situations.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect / Incomplete	(M, C) records the numbers of missionaries and cannibals on the left bank, but it omits the boat's location. The boat side is necessary to determine which crossings are possible.
Initial state	Incorrect / Incomplete	(3, 3) omits the boat location. The complete initial state is (3, 3, L).
Actions	Correct, but require conditions	The listed loads are the possible boat passenger combinations. An action is legal only if the boat is on the departure bank, enough people are present, and the resulting state satisfies the safety constraints.
Goal test	Incorrect	M = C is not the goal. The goal is to move all missionaries and cannibals to the right bank, so M = 0 and C = 0 on the left, with the boat on the right.
Legality check	Incorrect	The rule checks only the left bank. It must also check the right bank. On either bank, cannibals must not outnumber missionaries when at least one missionary is present.
Path cost	Incorrect	Counting the number of people carried does not match the objective of using the fewest crossings. Each crossing should have a cost of 1.
Note N1	Incorrect	The order is wrong. The agent should formulate a goal, formulate the problem, search for a solution and then execute the solution.
Note N2	Incorrect / Incomplete	There are 16 possible (M, C) combinations if boat location is omitted, of which 10 satisfy both bank-safety constraints. However, the complete state includes the boat side, giving 32 raw combinations and 20 safe states.

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

State representation	A state representation must include all information required to determine the available actions and the result of each action. Omitting the boat side makes the state incomplete.
Initial state	The initial state must specify every component of the complete state. The student included the people on the left bank but not the boat location.
Actions	The possible passenger loads are correct, but actions must be applied only when they are feasible from the current state and produce a legal result.
Goal test	The student confused equality between M and C with completion of the task. The goal must represent everyone reaching the right bank.
Legality check	Safety must be checked on both banks. Checking only the left bank may accept a state in which cannibals outnumber missionaries on the right bank.
Path cost	The student confused the number of people transported with the number of crossings. The stated objective is to minimise crossings, so each crossing has unit cost.
Note N1	A problem-solving agent must establish what it wants to achieve and define the problem before searching for a solution. Execution follows the search.
Note N2	The student counted states using an incomplete representation. Including the boat side doubles the raw state combinations from 16 to 32 and doubles the number of safe states from 10 to 20.

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	**Corrected formulation**	**Justification**
State representation	Represent a state as **(M, C, B)**, where M is the number of missionaries on the left bank, C is the number of cannibals on the left bank, and B is the boat side, L or R.	This representation includes the people's distribution and the boat location, which determine available actions.
Initial state	**(3, 3, L)**	All three missionaries and all three cannibals begin on the left bank, and the boat starts on the left.
Actions	Move 1M, 2M, 1C, 2C or 1M + 1C across the river, in the direction indicated by the boat's current side.	The boat must carry one or two people, cannot cross empty, and must not produce an unsafe state.
Goal test	**M = 0 and C = 0 and B = R.**	All people must be on the right bank, with the boat also on the right after the final crossing.
Legality check	For each bank, either the number of missionaries is zero or the number of missionaries is at least the number of cannibals. Also require $0 \leq M, C \leq 3$.	The safety condition must hold on both banks. The right-bank counts are $3 - M$ missionaries and $3 - C$ cannibals.
Path cost	Assign a cost of **1 per crossing**. The path cost is the total number of crossings.	This directly represents the objective of reaching the goal using the fewest crossings.
Note N1	**Goal formulation → problem formulation → search → execution.**	The agent first defines the desired outcome, then represents the problem, searches for a solution and executes it.
Note N2	There are **32 raw states** using (M, C, B), since 4 possible values of M × 4 values of C × 2 boat sides = 32. Of these, **20 satisfy the bank-safety constraints**.	Including the boat side doubles the number of state combinations compared with the incomplete representation.

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

(ii) Give one state that the student's legality check accepts although it is actually illegal.
(iii) Using your corrected path cost, state the cost of an optimal solution.

(i) Boat loads available from the initial state

Initial state: **(3, 3, L)**

The boat travels from the left bank to the right bank. The resulting state is written as (M, C, boat side).

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M	(2, 3, R)	**Illegal.** On the left bank, 2 missionaries and 3 cannibals remain. Cannibals outnumber missionaries.
2M	(1, 3, R)	**Illegal.** On the left bank, 1 missionary and 3 cannibals remain. Cannibals outnumber missionaries.
1C	(3, 2, R)	**Legal.** The left bank has 3 missionaries and 2 cannibals. The right bank has 0 missionaries and 1 cannibal, which is safe because there are no missionaries on that bank.
2C	(3, 1, R)	**Legal.** The left bank has 3 missionaries and 1 cannibal. The right bank has 0 missionaries and 2 cannibals, which is safe because there are no missionaries on that bank.
1M + 1C	(2, 2, R)	**Legal.** The left bank has 2 missionaries and 2 cannibals, and the right bank has 1 missionary and 1 cannibal. Neither bank violates the safety rule.

(ii) Give one state accepted by the student's legality check that is actually illegal.

Example: **(M, C, B) = (2, 1, R)**

The student's check accepts this state because $C \leq M$ on the left bank ($1 \leq 2$).

However, the right bank contains 1 missionary and 2 cannibals:

- Right-bank missionaries = $3 - 2 = 1$
- Right-bank cannibals = $3 - 1 = 2$

Cannibals outnumber missionaries on the right bank, so the state is **illegal**.

(iii) Using the corrected path cost, state the cost of an optimal solution.

The path cost is **11 crossings**, because each crossing costs 1 and the optimal solution requires 11 crossings.

One optimal sequence of crossings is:

1. Two cannibals cross to the right.
2. One cannibal returns to the left.
3. Two cannibals cross to the right.
4. One cannibal returns to the left.
5. Two missionaries cross to the right.
6. One missionary and one cannibal return to the left.
7. Two missionaries cross to the right.
8. One cannibal returns to the left.
9. Two cannibals cross to the right.
10. One cannibal returns to the left.
11. Two cannibals cross to the right.

This sequence preserves the safety constraint on both banks and moves everyone to the right bank in **11 crossings**.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3 and subsequent trace	The queue order after expanding D is incorrect. The existing node A must remain at the front of the FIFO queue, giving [A, B, E, G], not [B, E, G, A]. The later dequeue order and reported count are consequently wrong.
2	DFS, iteration 1 and subsequent trace	The stack after expanding S is shown as [D, C]. Because the stack top is on the left and the alphabetically first child must be popped first, the correct stack is [C, D]. The subsequent DFS trace and result are therefore wrong.
3	UCS, iteration 4	After expanding E with g(E) = 5, the path from E to G has cost 5 + 3 = 8. The frontier entry G:9(D) must be replaced with G:8(E).
4	UCS, iterations 5–7 and result	The cheaper route to G is not carried forward. UCS should remove G with cost 8, not cost 9. The UCS result is therefore not optimal as reported. This error follows from the missed update in iteration 4.
5	Remark R1	The claim that BFS does not need an explored set is incorrect. Graph search uses an explored set to prevent repeated expansion and to control cycles.
6	Remark R2	The claim that all three strategies are optimal because they returned the same path and cost is incorrect. The corrected DFS path differs, and BFS does not guarantee the minimum weighted path cost. UCS finds the least-cost path when edge costs are non-negative.
7	Remark R3	Removing the fewest nodes from the frontier does not automatically make DFS the best strategy. The appropriate strategy depends on the objective, solution quality and search characteristics.

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	**BFS uses a FIFO queue.** Nodes already in the queue retain their order. When D is expanded, A is already waiting in the queue, so new neighbours B, E and G must be placed after A. Incorrect queue ordering changes the subsequent search trace.
2	**DFS uses a LIFO stack.** With the top on the left, the alphabetically first child must be popped first. The children must be pushed in an order that leaves C at the top, resulting in [C, D].
3	**UCS maintains the lowest known path cost to frontier nodes.** When a cheaper path to a frontier node is discovered, its cost and parent must be updated. The route S–D–E–G costs 8, which is less than the previous route to G costing 9.
4	The missed cost update in iteration 4 was carried forward into later UCS iterations. UCS terminates with the goal when the lowest-cost goal node is removed from the priority queue, so G should be removed at cost 8.
5	In graph search, cycles and multiple paths can lead to repeated visits. An explored set records expanded nodes and prevents unnecessary re-expansion.

6	BFS and DFS search according to depth or traversal order, not weighted path cost. UCS considers cumulative path cost. A valid path found by a strategy is not necessarily the least-cost path.
7	Search efficiency must be evaluated against the problem's objective. A smaller number of frontier removals does not guarantee a lower path cost, completeness or optimality.

(c) Write the corrected traces and justify each correction.

(i) BFS: corrected queue after iteration 3 and dequeue order

Queue after iteration 3: `[A, B, E, G]`

Order of dequeued nodes: `S → C → D → A → B → E → G`

The corrected BFS trace follows. The goal test is applied when G is removed from the queue.

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[]	{S, C, D, A, B, E}

Justification: BFS follows FIFO order. At iteration 3, A was already in the queue before D was expanded. Therefore, A remains first, and the newly discovered nodes B, E and G are appended after it. When G is removed from the queue, the goal test succeeds.

(ii) DFS: corrected trace

The stack's top is on the left. Children are pushed so that the alphabetically first child is popped first.

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

Justification: From S, C is popped before D. From C, A is explored before S, which is already explored. From A, B is added because C is already explored. From B, G is added because A and D are already explored or on the stack. The goal test succeeds when G is popped.

(iii) UCS: corrected trace

The priority queue is ordered by cumulative path cost. Ties are broken alphabetically. Each entry is written as `Node:g(parent)`.

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)

5	B	7	G:8(E), A:9(C)
6	G (goal)	8	—

Justification of the corrections:

- In iteration 1, C and D both have cost 3, so the alphabetical tie-break places C first.
- Expanding C discovers A with cost 3 + 6 = 9.
- Expanding D discovers E with cost 3 + 2 = 5, B with cost 3 + 4 = 7, and G with cost 3 + 6 = 9.
- Expanding E discovers a cheaper path to G with cost 5 + 3 = 8. Therefore, the previous entry G:9(D) is replaced with G:8(E).
- Expanding B does not improve the existing frontier costs of G or A.
- G is removed at cost 8, which is the optimal path cost.

(d) Apply the corrected traces.

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S → D → G	9	7	Not guaranteed to be optimal by weighted path cost.
DFS	S → C → A → B → G	14	5	No. The path is not the least-cost path.
UCS	S → D → E → G	8	6	Yes. UCS finds the least-cost path for this graph with non-negative edge costs.

Path-cost calculations:

- **BFS:** S → D → G = 3 + 6 = 9.
- **DFS:** S → C → A → B → G = 3 + 6 + 2 + 3 = 14.
- **UCS:** S → D → E → G = 3 + 2 + 3 = 8.

(ii) Explain why an explored set is needed in graph search, using iteration 3 of corrected BFS.

At iteration 3, BFS expands D. Because the graph is undirected, D has S as one of its neighbours. However, S has already been expanded and is in the explored set `{S, C, D}`. The algorithm must not add S to the queue again.

The explored set prevents already expanded nodes from being repeatedly visited, reduces unnecessary work and helps avoid endless cycling in graphs. BFS must also avoid adding nodes that are already waiting in the queue; for example, A is already in the queue when D is expanded, so it must not be duplicated.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1	Task A, Step 3	After the final Suck action, the belief state is shown as {S6, S8}. Suck cleans the

		current square. Since the agent is on the right, S6 becomes S8, while S8 remains S8. The correct final belief state is {S8}.
2	Task B, Step 0 – after percept	The updated belief state is shown as {S1, S2, S3, S6}. The percept [Left, Dirty] must retain only states in which the agent is on the left and the left square is dirty: {S1, S3}.
3	Task B, Step 1 – after action	The predicted state set {S4, S5, S7, S8} is incorrect. Suck is applied to the corrected belief state {S1, S3}, producing {S5, S7}.
4	Task B, Step 2 – after action	The predicted set {S6, S7} is incorrect. Moving Right from {S5, S7} produces {S6, S8}. The updated belief state after the percept [Right, Dirty] is correctly {S6}.
5	Conclusion C1	A final belief state containing multiple states does not mean a sensorless problem cannot be solved. In Task A, the corrected final belief state is the singleton {S8}, which guarantees the goal.
6	Conclusion C2	The belief state {S6} does identify the agent's location and the current square's status, but the goal has **not** been reached because the right square is dirty.
7	Conclusion C3	There are not only 8 possible belief states. With 8 physical states, there are $2^8 = 256$ subsets, including the empty set, or 255 non-empty belief states.

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	The student did not apply the Suck transition correctly. Suck cleans the square currently occupied by the agent, so a dirty right square becomes clean while an already clean right square remains clean.
2	A percept update must filter the predicted belief state to retain only physical states consistent with the observation. [Left, Dirty] means the agent is at L and the left square is D, which applies only to S1 and S3.
3	The action prediction was calculated from an incorrect belief state. The action must be applied to each state in the corrected current belief state, and duplicate resulting states are represented only once.
4	Moving Right changes the agent's location but does not change the dirt status. From S5 and S7, the resulting states are S6 and S8. The subsequent percept filters out S8 because the right square is clean there.
5	The student confused uncertainty about the exact physical state with inability to achieve a goal. A sensorless plan can succeed when every state in the final belief state satisfies the goal, even if uncertainty remains.
6	The student confused knowing the current location and local status with satisfying the goal. A singleton belief state may still describe a world in which one square is dirty.
7	A belief state is any subset of the set of physical states, not just a single physical state. The number of subsets grows exponentially with the number of physical states.

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: corrected belief states

Initial belief state: `{S1, S3, S5, S7}`. The agent knows it is in the left square but does not know the dirt status of both squares.

Step	Action	Belief state after action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

Justification: Suck cleans the left square, giving S5 and S7. Moving Right changes these to S6 and S8. The final Suck cleans the right square in S6, producing S8; S8 remains unchanged because both squares are already clean.

(ii) Task B: corrected belief states after each action and percept

The initial belief state contains all eight physical states. The percept [Left, Dirty] filters this set to {S1, S3}.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

Justification:

- **Step 0:** [Left, Dirty] means the agent is in the left square and that square is dirty. Only S1 and S3 satisfy this percept.
- **Step 1:** Suck cleans the left square in both S1 and S3, resulting in S5 and S7. The percept [Left, Clean] is consistent with both states, so the belief state remains {S5, S7}.
- **Step 2:** Right moves the agent from the left square to the right square, producing S6 and S8. The percept [Right, Dirty] retains S6 and removes S8 because the right square is clean in S8.

(iii) Corrected conclusions C1 and C2, with justification

C1 – Corrected conclusion:

Sensorless problems can be solved when a plan guarantees that every physical state in the final belief state satisfies the goal. In Task A, the final belief state is `{S8}`, representing the agent in the right square with both squares clean. Therefore, the plan `[Suck, Right, Suck]` guarantees the goal.

Justification: A solution in belief-state search is a plan that reaches a belief state in which all possible physical states satisfy the goal. Sensors are not always necessary to guarantee a solution.

C2 – Corrected conclusion:

The corrected final belief state in Task B is `{S6}`. The agent knows it is in the right square, and the right square is dirty while the left square is clean. However, **the goal has not been reached**, because both squares must be clean.

Justification: A singleton belief state represents certainty about the physical state, but certainty does not mean the goal condition is satisfied. In S6 = (R, C, D), the right square remains dirty.

(d) Apply the corrected concepts.

(i) How many different belief states are possible with 8 physical states? Why is belief-state search harder?

A belief state is a subset of the set of physical states. For 8 physical states, the number of possible subsets is:

[
2^8 = 256
]

This includes the empty set. If only non-empty belief states are considered, the number is:

[
2^8 - 1 = 255
]

Therefore, there are **256 possible belief states in total, or 255 non-empty belief states**.

Belief-state search is harder because the number of possible belief states grows exponentially with the number of physical states. A physical-state search operates on individual states, while a belief-state search may need to consider many possible combinations of states. This increases the potential search space and the computation required.

(ii) Apply the plan [Right, Suck] to the initial belief state {S1, S3, S5, S7}. State whether it is a solution.

Step	Action	Belief state after action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}

Solution? No.

Why? After Right, the agent may be in any of S2, S4, S6 or S8. Applying Suck cleans the right square wherever it is dirty, producing the final belief state `{S4, S8}`.

- S4 = (R, D, C): the left square is still dirty.
- S8 = (R, C, C): both squares are clean.

Because S4 is still a possible physical state and does not satisfy the goal, the plan does not guarantee that both squares will be clean. Hence, `[Right, Suck]` is **not a solution**.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____